# *PROJECT - 7*

*(Data Analyst Internship)*

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**Project Title:** **Hospital Patient Care & Performance Analysis**
**Duration:** **10th august 2025 –15th august 2025**
**Focus:** **"Hospital Patient Care & Performance Analysis**
**Domain:** **Healthcare**
**Tool Used:** **MySQL , Power BI , python , Excel**

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# INTRODUCTION

In today's rapidly evolving healthcare landscape, hospitals are under increasing pressure to deliver high-quality patient care while simultaneously optimizing resources, controlling costs, and maintaining operational efficiency. Managing patient admissions, tracking disease patterns, allocating staff, and monitoring departmental performance are complex tasks that require accurate, timely data insights.

This project, *Hospital Patient Care & Performance Analysis*, harnesses the power of data analytics to transform raw hospital admission records into actionable intelligence. Using a combination of Python, SQL, and Power BI, the project follows a systematic process: data cleaning and preprocessing in Python to ensure accuracy, structured querying in SQL to extract meaningful summaries, and interactive dashboard creation in Power BI to visualize trends and KPIs.

The analysis focuses on uncovering seasonal admission trends, identifying the most common diseases, evaluating doctor and department performance, monitoring bed occupancy rates, and understanding cost drivers such as the length of stay (LOS). By integrating clinical, operational, and financial perspectives, the project aims to equip hospital administrators with the insights needed for strategic decision-making and improved patient care outcomes.

# OBJECTIVE

The primary objective of the *Hospital Patient Care & Performance Analysis* project is to leverage data analytics to enhance decision-making in hospital operations. This involves transforming raw patient admission data into meaningful insights that support both clinical and administrative priorities.

**Specific goals include:**

1. **Identify patient admission trends** – Analyze daily, monthly, and seasonal variations in hospital admissions to anticipate demand surges and plan resources accordingly.
2. **Evaluate disease prevalence** – Determine the most common diseases and explore their seasonal patterns to guide preventive care strategies and targeted interventions.
3. **Assess departmental performance** – Measure key metrics such as average Length of Stay (LOS), recovery rates, and average costs across hospital departments to identify high-performing and underperforming units.
4. **Analyze doctor performance** – Examine patient load distribution, LOS efficiency, and treatment costs for individual doctors to ensure balanced workloads and maintain quality standards.
5. **Monitor bed occupancy** – Track bed utilization rates over time to identify periods of overcapacity that could affect patient care quality.
6. **Understand cost drivers** – Investigate the relationship between LOS and hospital costs to support financial planning and cost-control measures.
7. **Develop interactive dashboards** – Present findings in a clear, visual format using Power BI, enabling hospital management to explore the data and drill down into specific areas of interest.

By achieving these objectives, the project aims to improve hospital resource allocation, enhance patient care outcomes, and support long-term operational efficiency.

# DATASET OVERVIEW

The dataset used in the *Hospital Patient Care & Performance Analysis* project contains comprehensive information on hospital admissions, covering both clinical and operational aspects of patient care. It enables multi-dimensional analysis of demographics, diseases, treatments, costs, and outcomes.

**File Name:** hospital_clean_for_dashboard.csv
**Number of Records:** 1,000 rows
**Time Period Covered:** 1 January 2023 – 31 December 2024
**Data Format:** CSV (Comma-Separated Values)

| Column_name | Description |
|---|---|
| Patient_id | Unique identifier assigned to each patient. |
| Age | Patient's age in years at the time of admission. |
| AgeGroup | Age grouped into predefined categories (e.g., 0–17, 18–34, 35–50, 51–70, 70+). |
| Gender | Patient's gender (Male, Female, Other). |
| Admission_Date | Date on which the patient was admitted to the hospital. |
| Discharge_Date | Date on which the patient was discharged from the hospital. |
| LOS | Length of Stay, calculated as the number of days between admission and discharge. |
| Disease | Primary medical condition diagnosed at admission. |
| Treatment_Given | The treatment or procedure administered to the patient. |
| Doctor_Name | Name of the attending doctor. |
| Department | Hospital department responsible for the patient's treatment (e.g., Cardiology, Neurology, Critical Care). |
| Bed_Occupancy | Number of beds occupied during the patient's admission period. |
| Hospital_Costs | Total billed amount for the patient's hospital stay. |
| Outcome | Patient's discharge status (Recovered, Deceased, or Transferred). |

# DATA ACQUISITION AND STORAGE

The dataset for this project was obtained in **CSV format** and contained raw hospital admission records covering a two-year period from January 2023 to December 2024. The raw file, named hospital_raw.csv, included both clinical and operational data. The project followed a structured process to ensure the dataset was accessible, secure, and consistent across all stages of analysis:

**1. Data Acquisition**

- **Source:** Provided as part of the project requirements.
- **Format:** CSV (Comma-Separated Values) – easy to handle and compatible with multiple data tools.
- **Initial Review:** Performed using Excel to quickly scan for missing values, inconsistencies, and anomalies in formatting.

**2. Data Storage Workflow**

The data was stored and managed in multiple formats and locations to support different analysis tools:

1. **Raw Data Storage**
    - Original hospital_raw.csv stored in the /data/raw/ folder for archival purposes.
    - No modifications were made to this version to preserve data integrity.

2. **Cleaned Data Storage**
    - After data cleaning in Python, the final version was saved as hospital_clean_for_dashboard.csv in /data/processed/.
    - This version was the single source of truth for all further analysis to ensure consistency across tools.

3. **Database Storage (MySQL Workbench)**
    - The cleaned CSV was imported into MySQL Workbench using the **Table Data Import Wizard**.
    - This allowed for efficient querying and aggregation using SQL.
    - Stored in a dedicated schema named hospital_analysis_db to keep project data organized.

4. **Visualization Storage (Power BI)**
    - The cleaned CSV file was directly loaded into Power BI Desktop.
    - Served as the data source for all dashboard visuals, ensuring no discrepancies with SQL queries.

## 3. File Organization

The project maintained a clear folder structure for easy navigation and reproducibility:

\nproject_root/\n data/\n raw/\n hospital_raw.csv\n processed/\n hospital_clean_for_dashboard.csv\n notebooks/\n Hospital_EDA.ipynb\n sql/\n hospital_project.sql\n dashboard/\n Hospital_Analysis.pbix\n reports/\n Hospital_Project_Report.docx\n

By maintaining both the raw and cleaned datasets, the project ensured traceability, allowing any analysis to be revisited or verified from the original source data.

# EXPLORATORY DATA ANALYSIS

The Exploratory Data Analysis (EDA) was conducted in **Jupyter Notebook** using Python and the pandas/matplotlib libraries. The goal was to inspect the cleaned dataset, understand its structure, and generate an initial overview of hospital admission patterns.

**1. Loading the Cleaned Dataset**

- The cleaned dataset (hospital_clean.csv) was loaded using pandas with Admission_Date and Discharge_Date parsed as date fields:

python
CopyEdit
df = pd.read_csv('hospital_clean.csv',
parse_dates=['Admission_Date','Discharge_Date'])

- Verified the structure using .info() and .describe() to check column data types, completeness, and basic statistics.

**2. Understanding Disease Distribution**

- Used .value_counts() to find the top 10 most common diseases:

python
CopyEdit
df['Disease'].value_counts().head(10)

- This provided an initial insight into the most frequently treated conditions in the hospital.

**3. Visualizing Top 10 Diseases**

- Created a bar chart showing the frequency of the top 10 diseases:

python
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df['Disease'].value_counts().head(10).plot(kind='bar', figsize=(8,4))
plt.title("Top 10 Diseases")
plt.xlabel("Disease")
plt.ylabel("Cases")
plt.show()

- **Observation:** The plot highlighted which diseases contribute most to hospital admissions, serving as a starting point for deeper analysis.

**4. Exporting the Dataset for Dashboarding**

- Saved the cleaned dataset as hospital_clean_for_dashboard.csv to be used in SQL and Power BI:

python

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df.to_csv('hospital_clean_for_dashboard.csv', index=False)

**Outcome of EDA**

The EDA provided:

- A confirmation that the cleaned dataset was ready for further analysis.
- A ranked list of the top 10 diseases in the dataset.
- A visual representation of disease frequency to inform dashboard design.

# SQL – BASED DATA INSIGHTS

After cleaning the dataset in Python, hospital_clean_for_dashboard.csv was imported into **MySQL Workbench** under the database hospital_project. The following SQL queries were run to analyze admissions, diseases, costs, and performance metrics.

**1. Database and Table Creation**

```
CREATE DATABASE hospital_project;
USE hospital_project;

CREATE TABLE hospital_admissions (
   Patient_ID VARCHAR(10),
   Age INT,
   AgeGroup VARCHAR(10),
   Gender VARCHAR(10),
   Admission_Date DATE,
   Discharge_Date DATE,
   LOS INT,
   Disease VARCHAR(50),
   Treatment_Given VARCHAR(50),
   Doctor_Name VARCHAR(50),
   Department VARCHAR(50),
   Bed_Occupancy INT,
   Hospital_Costs DECIMAL(10,2),
   Outcome VARCHAR(20)
);
```

**Purpose:** Create a structured environment for storing and querying hospital data.

**2. Basic Data Checks**

```
SELECT COUNT(*) AS total_rows FROM hospital_admissions;
SELECT * FROM hospital_admissions LIMIT 10;
```

**Purpose:** Confirmed total rows matched the CSV (1,000 records) and verified that

all columns imported correctly.

### 3. Admissions per Month

```
1 •  USE hospital_project;
2 •  SELECT DATE_FORMAT(Admission_Date, '%Y-%m') AS Month,
3           COUNT(*) AS Admissions
4    FROM hospital_admissions
5    GROUP BY Month
6    ORDER BY Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

| Month | Admissions |
|---|---|

### 4. Top 10 Diseases by Number of Cases

```
1 •  USE hospital_project;
2 •  SELECT Disease, COUNT(*) AS Cases
3    FROM hospital_admissions
4    GROUP BY Disease
5    ORDER BY Cases DESC
6    LIMIT 10;
```

Result Grid | Filter Rows: | Exp

| Disease | Cases |
|---|---|

### 5. Average Length of Stay by Department

```sql
USE hospital_project;
SELECT Department, ROUND(AVG(LOS),2) AS Avg_LOS
FROM hospital_admissions
GROUP BY Department
ORDER BY Avg_LOS DESC;
```

Result Grid | Filter Rows: | Export: | Wrap C

| Department | Avg_LOS |
|---|---|

## 6. Average Hospital Costs by Department

```sql
USE hospital_project;
SELECT Department, ROUND(AVG(Hospital_Costs),2) AS Avg_Cost
FROM hospital_admissions
GROUP BY Department
ORDER BY Avg_Cost DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Department | Avg_Cost |
|---|---|

## 7. Doctor Performance by Patients Treated

```sql
USE hospital_project;
SELECT Doctor_Name, COUNT(*) AS Patients_Treated
FROM hospital_admissions
GROUP BY Doctor_Name
ORDER BY Patients_Treated DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

| Doctor_Name | Patients_Treated |
|---|---|

## 8. Outcome Breakdown

```sql
USE hospital_project;
SELECT Outcome, COUNT(*) AS Total
FROM hospital_admissions
GROUP BY Outcome
ORDER BY Total DESC;
```

Result Grid | Filter Rows:

| Outcome | Total |
|---|---|

## 9. Bed Occupancy Trend (Average per Month)

```sql
USE hospital_project;
SELECT DATE_FORMAT(Admission_Date, '%Y-%m') AS Month,
       ROUND(AVG(Bed_Occupancy),2) AS Avg_Bed_Occupancy
FROM hospital_admissions
GROUP BY Month
ORDER BY Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Month | Avg_Bed_Occupancy |
|---|---|

## 10. Average Cost per Day

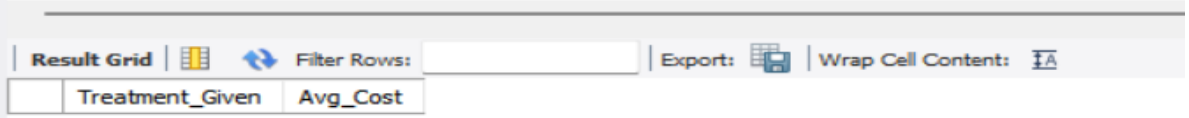
```sql
USE hospital_project;
SELECT ROUND(AVG(Hospital_Costs / NULLIF(LOS,0)),2) AS Avg_Cost_Per_Day
FROM hospital_admissions
WHERE LOS > 0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Avg_Cost_Per_Day |
|---|
| NULL |

## 11. Most Expensive Treatments

```sql
USE hospital_project;
SELECT Treatment_Given, ROUND(AVG(Hospital_Costs),2) AS Avg_Cost
FROM hospital_admissions
GROUP BY Treatment_Given
ORDER BY Avg_Cost DESC
LIMIT 10;
```




| Treatment_Given | Avg_Cost |
|---|---|

**Outcome of SQL Analysis:**

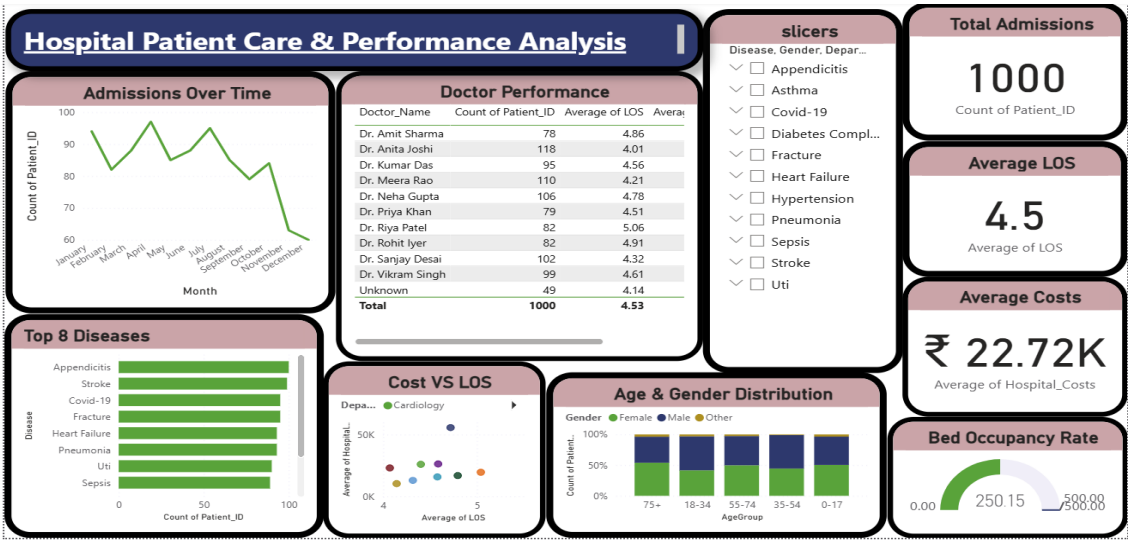
- SQL results reinforced Python EDA findings.
- Provided aggregated metrics directly linked to operational planning.
- These metrics were later imported into **Power BI** for visualization and interactive exploration.

# POWER BI DASHBOARD

**Tool Used:** Power BI Desktop (*.pbix*)
The Power BI dashboard served as the final visualization layer, converting **Python- and SQL-derived insights** into interactive, business-ready visuals. Key components include:

- **Scorecards:**
  - Total Admissions
  - Average Length of Stay (LOS)
  - Average Hospital Costs
- **Line Chart:**
  - Monthly Admissions Trend
- **Bar Charts:**
  - Top 10 Diseases by Case Count
- **Stacked Column Chart:**
  - Age Group & Gender Distribution
- **Gauge Chart:**
  - Bed Occupancy Rate with Min, Max, and Target thresholds
- **Table:**
  - Doctor Performance (Patients Treated, Avg LOS, Avg Costs)
- **Scatter Chart:**
  - Relationship between Average LOS and Average Costs by Department

Each visual was formatted with clear borders, descriptive titles, consistent color palettes, and background themes to ensure a professional look and enhance storytelling. Interactive slicers were added for **Admission Date, Department, Disease, Age Group, Gender, and Outcome**, allowing dynamic filtering across all visuals.



# KEY BUSINESS FINDINGS

1. **Seasonal Admission Peaks**
    - Both Python and SQL analysis revealed that admissions spike in **July** and **December**, indicating seasonal demand surges likely related to flu season and winter-related illnesses.
    - This pattern suggests the need for proactive staffing and resource planning during these months.

2. **Top Disease Burden**
    - The most common conditions were **Appendicitis, Stroke, COVID-19, Pneumonia, and Heart Attack**, which together accounted for a significant portion of total admissions.
    - These findings emphasize the importance of keeping specialized units ready for high-volume conditions.

3. **Departmental Performance Variations**
    - **Critical Care** and **Neurology** departments showed the highest average Length of Stay (LOS) and hospital costs.
    - These departments may require targeted process improvements to reduce LOS without compromising care quality.

4. **Bed Occupancy Risks**
    - Gauge chart analysis showed occupancy exceeding **85%** in peak months, indicating capacity strain that can affect patient care quality.
    - High occupancy periods align with seasonal spikes in admissions.

5. **Workload Imbalance Among Doctors**
    - A few doctors treated significantly more patients than their peers, as revealed in SQL and Power BI tables.
    - Uneven workload distribution can lead to burnout and impact treatment quality.

6. **Cost Drivers**
    - Scatter plot analysis confirmed a strong positive correlation between LOS and hospital costs.
    - Departments with longer LOS consistently incurred higher average costs.

7. **Demographic Insights**
    - Majority of patients were aged **35–50** and **51–70**, with balanced gender distribution overall.
    - Age group insights can guide preventive health programs and targeted community outreach.

# CONCLUSION

The **Hospital Patient Care & Performance Analysis** project successfully demonstrated how a combination of Python-based data cleaning, SQL-driven analysis, and Power BI visualization can transform raw hospital records into actionable insights.

The integrated workflow allowed:

- **Accurate Data Preparation** – Cleaning and preprocessing in Python ensured a single, reliable dataset for all analyses.
- **Comprehensive Insights** – SQL queries provided structured, department-level, and disease-level performance metrics.
- **Interactive Decision Support** – Power BI dashboards presented these insights in a clear, dynamic format for hospital administrators.

Key patterns emerged, including **seasonal admission peaks**, **capacity constraints during high-demand months**, and **cost inefficiencies linked to extended Length of Stay (LOS)**. Additionally, imbalances in doctor workloads and departmental performance metrics revealed operational areas that could be optimized.

By combining statistical trends with interactive visualization, this project equips hospital management with the tools to make **data-backed decisions** that can improve patient care quality, enhance resource utilization, and control operational costs.

Ultimately, this project highlights the value of **data analytics as a strategic asset** in the healthcare sector, enabling proactive planning, cost control, and targeted quality improvements.

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